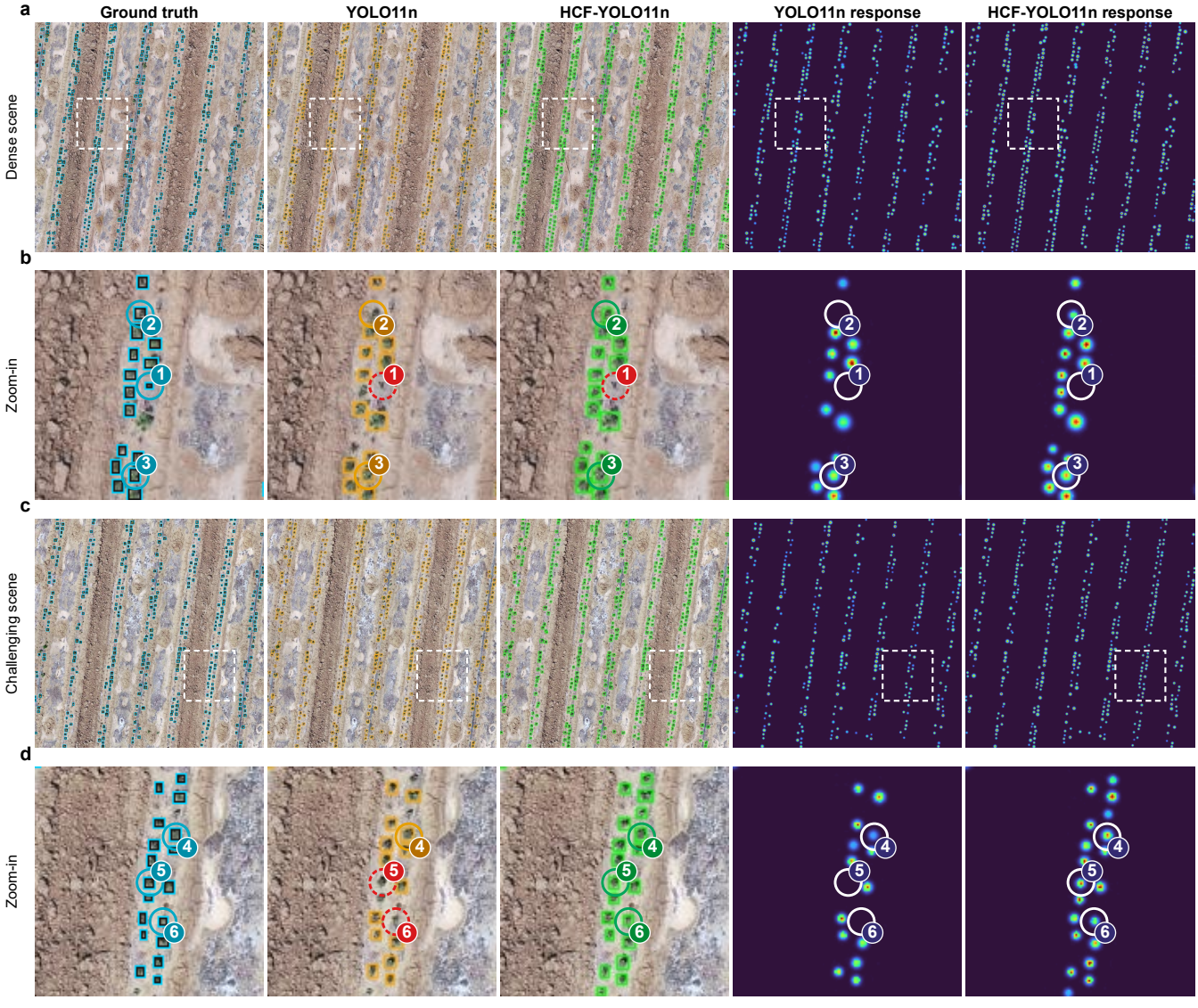

a
Ground truth
YOLO11n
HCF-YOLO11n
YOLO11n response
HCF-YOLO11n response
Dense scene
b
Zoom-in
c
Challenging scene
d
Zoom-in
GT: cyan
YOLO11n: orange
HCF-YOLO11n: green
1 shared miss; 2 stronger HCF; 3-4 shared TP; 5-6 recovered by HCF